

Jiawei Chen

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[Personal Website](#) | [Google Scholar](#) | [ORCID](#)

EDUCATION

- **Ph.D.** in Mechanics (Main advisor: Prof. Song Fu; Associate advisor: Prof. Yufei Zhang)
Tsinghua University, Beijing, China 08/2020 – 06/2024
Dissertation: *Numerical Simulation Study on Ice Accretion of Wings*
Ranked 1st in cohort
- **M.S.E.** in Power Engineering (Main advisor: Prof. Songtao Wang)
Harbin Institute of Technology, Harbin, China 08/2017 – 06/2019
Ranked 1st in cohort
- **B.S.E.** in Energy and Power Engineering
Harbin Engineering University, Harbin, China 08/2013 – 06/2017
Ranked 1st in major

PROFESSIONAL EXPERIENCE

- **Iowa State University**, Department of Aerospace Engineering
Postdoctoral Research Associate (Advisor: Prof. Ping He) 09/2025 – Present
- **University of Texas at Arlington**, Department of Mathematics
Postdoctoral Research Associate (Advisor: Prof. Chaoqun Liu) 11/2024 – 08/2025

RESEARCH INTERESTS

- Aerospace Aerodynamics and Aircraft Icing
- Turbulence Modeling and Simulation (Transition, RANS, LES)
- Convective Heat Transfer and Multiphase Flows (Ice Accretion)
- Data-Driven Aerodynamic Modeling for Aerospace Design and Control
- Agentic AI & Large Language Models for Fluid Dynamics

RESEARCH PROJECTS

- **An Integrated Numerical and Experimental Study of Wind-Driven Water Film Flow Dynamics Pertinent to Wind Turbine Icing Phenomena.** National Science Foundation (NSF) Award [2415347](#). Total award: \$620,900. 12/2024–11/2027. PI: Ping He; Co-PI: Hui Hu (Iowa State University).
Role: Postdoctoral Research Associate (2025–present). Developed a transformer–diffusion generative model to predict the unsteady flow evolution over iced airfoils, and applied the PhysMiner agentic-AI framework to iced-wing flows, yielding new insights into their flow structures.
- **Development and Application of Liutex and Third Generation of Vortex Definition and Identification.** NSF EAGER Award [2300052](#), Fluid Dynamics Program. Total award: \$243,229. 01/2023–09/2025. PI: Chaoqun Liu (University of Texas at Arlington).
Role: Postdoctoral Research Associate (2024–2025). Developed Liutex-based subgrid-scale

models for LES and a relative Liutex vortex identification method.

RESEARCH PROPOSAL & TRAINING

- **Research Proposal: conceptualized, drafted and led one NSF proposal as PI**

A Large Language Model–Driven Agentic Framework for Automated Turbulence Model Development. National Science Foundation (NSF) Computational and Data-Enabled Science and Engineering (CDS&E). Total award: \$300K. 2027–2030. **PI: Jiawei Chen.** Co-PI: Prof. Ping He. (**under review**)

- **Grant-Writing Training & PI Eligibility**

Completed independent grant-writing training. Earned the Center for Communication Excellence (CCE) Grantsmanship Summer Intensive microcredential, Iowa State University (Badge Level 1: Finding Grants; Badge Level 2: Preparing Applications), covering funder identification, eligibility and competitiveness assessment, and proposal narrative development; produced a project Gantt chart and a response to a mock Funding Opportunity Announcement.

ISU-approved Principal Investigator, registered with NSF Research.gov. Completed Iowa State University’s Research Security Training, modeled after NSF, NIH, DoD, and DoE guidelines (export controls, disclosure policies, risk mitigation, and responsible international collaboration).

OPEN-SOURCE SOFTWARE

- **PhysMiner:** Agentic-AI framework for automated flow component analysis, integrating triple-decomposition flow analysis with LLM-driven reasoning

Status: Open to the public. Validated end-to-end on three cases: periodic-hill flow, an iced airfoil, and a propeller. For the periodic hill, it autonomously derived an improved subgrid-scale model with more accurate Reynolds-stress predictions. For the iced airfoil and propeller, it revealed flow structures that conventional methods fail to identify.

Reference: **Chen, J.,** Gao, H., & He, P. PhysMiner: An Agentic AI Framework for Automated Flow Component Analysis. *AIAA Journal*, 2026. under review

Code: <https://github.com/iDesign-Lab/PhysMiner>

- **JAX-LES:** Differentiable computational fluid dynamics solver built on JAX, enabling gradient-based aerodynamic design optimization and machine-learning-augmented turbulence modeling directly within the flow solver

Status: In active development. Validated on decaying isotropic turbulence, channel flow, periodic hill, and airfoil. Planned for open-source release.

Role: Lead developer; designed and implemented the solver.

Reference: **Chen, J.,** & He, P. (in preparation)

TEACHING AND MENTORING

- **Teaching Experience**

Teaching Assistant, Tsinghua University

Spring 2023

Advanced Turbulence Simulation, graduate course taught by Prof. Song Fu. Delivered selected lectures as substitute instructor, graded homework assignments, and held after-class office

hours for student questions.

- **Teaching Preparation**

Preparing Future Faculty (PFF) Program, Iowa State University Fall 2026

Selected participant; semester-long seminar (GRST 5850) covering course and job-material development, the teaching–research–service balance, and the promotion and tenure process, paired with biweekly meetings with a tenure-track faculty mentor.

Courses Prepared to Teach

Core/Undergraduate: Fluid Mechanics; Heat Transfer; Numerical Methods for Engineers; Introduction to Aerodynamics; Introduction to CFD; Senior Design (capstone support)

Graduate/Elective: Advanced Fluid Mechanics; Computational Fluid Dynamics; Turbulence Modeling and Simulation; Agentic AI and Generative AI for Fluid Dynamics

- **Mentoring**

Emran Hossen, Ph.D. Student in Mathematics, UT Arlington 2024 – 2025

Mentored him (graduated Spring 2025) on Computational Fluid Dynamics (CFD) and turbulence modeling for his doctoral dissertation.

Collaboratively worked on a Liutex-based subgrid model for Large Eddy Simulation (LES), leading to a publication and presentation at the AIAA SciTech 2025 Forum.

Mentee successfully secured a Postdoctoral Scholar position at Michigan Technological University.

PROFESSIONAL SERVICE

- **Journal Paper Reviewing**

AIAA Journal

Physics of Fluids

Aerospace Science and Technology

International Journal of Heat and Fluid Flow

- **Professional Membership**

American Physical Society (APS)

- **Seminar/Conference Services**

Reviewer for GPPS-TC-2026 conference papers: [GPPS Montreal 2026](#), Montreal, Canada, September 9–11, 2026

PUBLICATIONS

Summary: 14 peer-reviewed journal papers (9 first-author), 2 journal papers under review, and 3 conference papers.

Journal Papers Under Review

[S2] **J. Chen**, H. Gao, and P. He. PhysMiner: An Agentic AI Framework for Automated Flow Component Analysis. *AIAA Journal*, 2026. ([Under Review](#)) [arXiv:2607.04009](#).

[S1] **J. Chen**, Y. Yu, E. Hossen, and C. Liu. Near-Wall Scaling and Separation Prediction of a Rotation-Based Subgrid-Scale Stress Model. *Journal of Turbulence*, 2025. ([Under Review](#)) [arXiv:2507.20043](#).

Peer-Reviewed Journal Papers

- [J14] **J. Chen**, P. He, and H. Hu. Transformer–Diffusion Modeling of Unsteady Flow Over an Iced Airfoil. *AIAA Journal*, 2026. (In Press)
- [J13] Z. Li, L. Fang, **J. Chen**, A. Sharma, and P. He. Field Inversion Machine Learning for Time-Resolved Unsteady Flows in Airfoil Dynamic Stall. *AIAA Journal*, 2026. [doi:10.2514/1.J066648](https://doi.org/10.2514/1.J066648)
- [J12] **J. Chen**, M. Xiao, Z. Zhou, and Y. Zhang. Numerical investigation of discontinuous ice effects on swept wings. *International Journal of Heat and Fluid Flow*, 120, p. 110391, 2026. [doi:10.1016/j.ijheatfluidflow.2026.110391](https://doi.org/10.1016/j.ijheatfluidflow.2026.110391)
- [J11] **J. Chen**, Z. Zhou, M. Xiao, and Y. Zhang. Flow-separation simulations of continuous and discontinuous ice on swept wings. *Aerospace Science and Technology*, 174, p. 111891, 2026. [doi:10.1016/j.ast.2026.111891](https://doi.org/10.1016/j.ast.2026.111891)
- [J10] Z. Zhou, M. Xiao, **J. Chen**, L. Li, and Y. Zhang. Numerical Investigation of Sweep Effects on Turbulent Flow over Iced Wings. *AIAA Journal*, 64(2), pp. 823–839, 2026. [doi:10.2514/1.J065820](https://doi.org/10.2514/1.J065820)
- [J9] M. Xiao, Y. Wang, F. Rodach, B. Font, M. Kurz, P. Suárez, D. Zhou, F. Alcántara-Ávila, T. Zhu, J. Liu, R. Montalà, **J. Chen**, J. Rabault, O. Lehmkuhl, A. Beck, J. Larsson, R. Vinuesa, and S. Pirozzoli. SmartFlow: A CFD-solver-agnostic deep reinforcement learning framework for computational fluid dynamics on HPC platforms. *Computer Physics Communications*, 2025. (In Press) [arXiv:2508.00645](https://arxiv.org/abs/2508.00645).
- [J8] **J. Chen**, C. Yan, and Y. Zhang. Separation Prediction and Roughness Effect Simulation of Iced Airfoils Based on a Modified $k\text{-}\omega\text{-}\gamma$ Transition Model. *Aerospace Science and Technology*, 162, p. 110202, 2025. [doi:10.1016/j.ast.2025.110202](https://doi.org/10.1016/j.ast.2025.110202)
- [J7] **J. Chen**, Y. Zhang, and S. Fu. Ice Accretion Simulation Incorporating Roughness Effects and Separation Correction through a Transition Model. *Physics of Fluids*, 36(4), 2024. [doi:10.1063/5.0209004](https://doi.org/10.1063/5.0209004) (Editor’s Pick)
- [J6] **J. Chen**, Y. Zhang, and S. Fu. Numerical study of the effects of discontinuous ice on three-dimensional wings. *Physics of Fluids*, 36(7), 2024. [doi:10.1063/5.0217642](https://doi.org/10.1063/5.0217642)
- [J5] **J. Chen**, P. Yang, Y. Zhang, and S. Fu. Aerodynamic prediction and roughness implementation toward ice accretion simulation. *Physics of Fluids*, 36(1), 2024. [doi:10.1063/5.0184630](https://doi.org/10.1063/5.0184630)
- [J4] P. Yang, H. Li, **J. Chen**, Y. Duan, and Y. Zhang. Influence of Air Flow Field Computation on an Ice Accretion Simulation of Airfoils. *Aerospace Science and Technology*, 151, p. 109315, 2024. [doi:10.1016/j.ast.2024.109315](https://doi.org/10.1016/j.ast.2024.109315)
- [J3] **J. Chen**, Y. Zhang, and S. Fu. Roughness implementation and convective heat transfer coefficient computation toward ice accretion simulation. *Physics of Fluids*, 35(11), 2023. [doi:10.1063/5.0173847](https://doi.org/10.1063/5.0173847) (Featured Article)
- [J2] **J. Chen**, L. Cai, and S. Wang. Numerical Study on Effects of Different Excitation Forms on Forced Film Cooling Characteristics. *Journal of Propulsion Technology*, 41(4), pp. 860–867, 2020.

- [J1] S. Sun, S. Wang, S. Chen, C. Tao, L. Cai, and **J. Chen**. The impact of various forward sweep angles on the performance of an ultra-high-load low-reaction transonic compressor rotor. *Applied Thermal Engineering*, 150, pp. 953–966, 2019. [doi:10.1016/j.applthermaleng.2019.01.045](https://doi.org/10.1016/j.applthermaleng.2019.01.045)

Conference Papers

- [C3] **J. Chen**, Y. Yu, and C. Liu. A Relative Liutex Method for Vortex Identification. *Vortex 2025 Conference Proceedings*, 2025. (In Press) [arXiv:2512.23857](https://arxiv.org/abs/2512.23857).
- [C2] **J. Chen**, Y. Yu, and C. Liu. A Dynamic Subgrid-Scale Model Based on Liutex Theory for Wall-Bounded Turbulent Flows. *Vortex 2025 Conference Proceedings*, 2025. (In Press) [arXiv:2512.14953](https://arxiv.org/abs/2512.14953).
- [C1] E. Hossen, Y. Yu, C. Liu, **J. Chen**, Z. Liu, Y. Yan, V. Shah, and O. Alvarez. Liutex-Based Sub-Grid Model for Large Eddy Simulation of Turbulent Flow. *AIAA SciTech 2025 Forum*, AIAA 2025-0073, 2025. [doi:10.2514/6.2025-0073](https://doi.org/10.2514/6.2025-0073)

PRESENTATIONS

- [P6] **J. Chen**, H. Gao, and P. He. PhysMiner: An Agentic AI Framework for Automated Flow Component Analysis. *79th Annual Meeting of the APS Division of Fluid Dynamics (DFD)*, November 22, 2026. (Interactive Presentation, upcoming)
- [P5] **J. Chen**, Y. Yu, and C. Liu. A Relative Liutex Method for Vortex Identification. *Vortex 2025 Conference*, University of Texas at Arlington, Arlington, TX, December 2025.
- [P4] **J. Chen**, Y. Yu, and C. Liu. A Dynamic Subgrid-Scale Model Based on Liutex Theory for Wall-Bounded Turbulent Flows. *Vortex 2025 Conference*, University of Texas at Arlington, Arlington, TX, December 2025.
- [P3] **J. Chen**, Y. Zhang, and S. Fu. Implementation and Effects of Roughness in Aircraft Icing Simulation. *Civil Aviation Safety Thematic Academic Conference*, Guanghan, Sichuan, China, December 2023.
- [P2] **J. Chen**, Y. Zhang, and S. Fu. The Extended Transition Model for Convective Heat Transfer Coefficient Computation in Three-Dimensional Icing. *Joint Conference of APCATS, AJS&A & AAME 2023*, Jeju, Korea, October 11–14, 2023.
- [P1] **J. Chen**, Y. Zhang, and S. Fu. Key Issues in the Aircraft Icing Simulation Framework. *5th National Conference on Icing and Anti-/De-icing*, Changsha, China, May 2023.

HONORS AND AWARDS

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| • Outstanding Graduate, School of Aerospace Engineering, Tsinghua University | 2024 |
| • Editor’s Pick, <i>Physics of Fluids</i> (J7) | 2024 |
| • Featured Article, <i>Physics of Fluids</i> (J3) | 2023 |
| • First-Class Scholarship, School of Aerospace Engineering, Tsinghua University | 2021 |
| • First-Class Scholarship, Tsinghua University | 2021 |
| • National Scholarship, Ministry of Education of China | 2018 |
| • Innovation Scholarship (First Class), Ministry of Industry and Information Technology (MIIT) of China | 2018 |
| • First-Class Scholarship, Harbin Institute of Technology | 2018 |

- Top Ten Outstanding Students, Harbin Institute of Technology

2018

TECHNICAL SKILLS

- **Programming Languages:** Python (JAX), C++, Fortran, MATLAB
- **Software:** OpenFOAM, ANSYS, ParaView, Tecplot, Linux, Slurm
- **High-Performance Computing:** Texas Advanced Computing Center (TACC) Stampede3; Nova HPC Cluster (Open OnDemand), Iowa State University